Customer Churn Prediction

Assignment 2 — Flask app using the Assignment 1 model

Project summary

A Flask web application serves the tuned Assignment 1 Logistic Regression model through a customer form. The persisted scikit-learn pipeline applies the fitted imputation, scaling, and one-hot encoding. The form accepts the 19 model features and returns a plain-language estimate and churn probability.

Model and evaluation

Item	Result
Model	Logistic Regression; tuned C = 30
Training-selected threshold	0.55
Five-fold CV accuracy	80.37%
Stratified holdout accuracy	79.77%
Holdout churn precision / recall / F1	67.05% / 46.79% / 55.12%
Holdout ROC AUC	0.8409
Majority-class baseline	73.46%

The requested accuracy above 85% was not achieved; results are reported as measured. The holdout set was not used to select the model or decision threshold.

Project structure

```
project/
  app.py
  model.joblib
  requirements.txt
  churn_prediction.ipynb
  templates/index.html
  static/style.css
  tests/test_app.py
```

Implementation and local verification

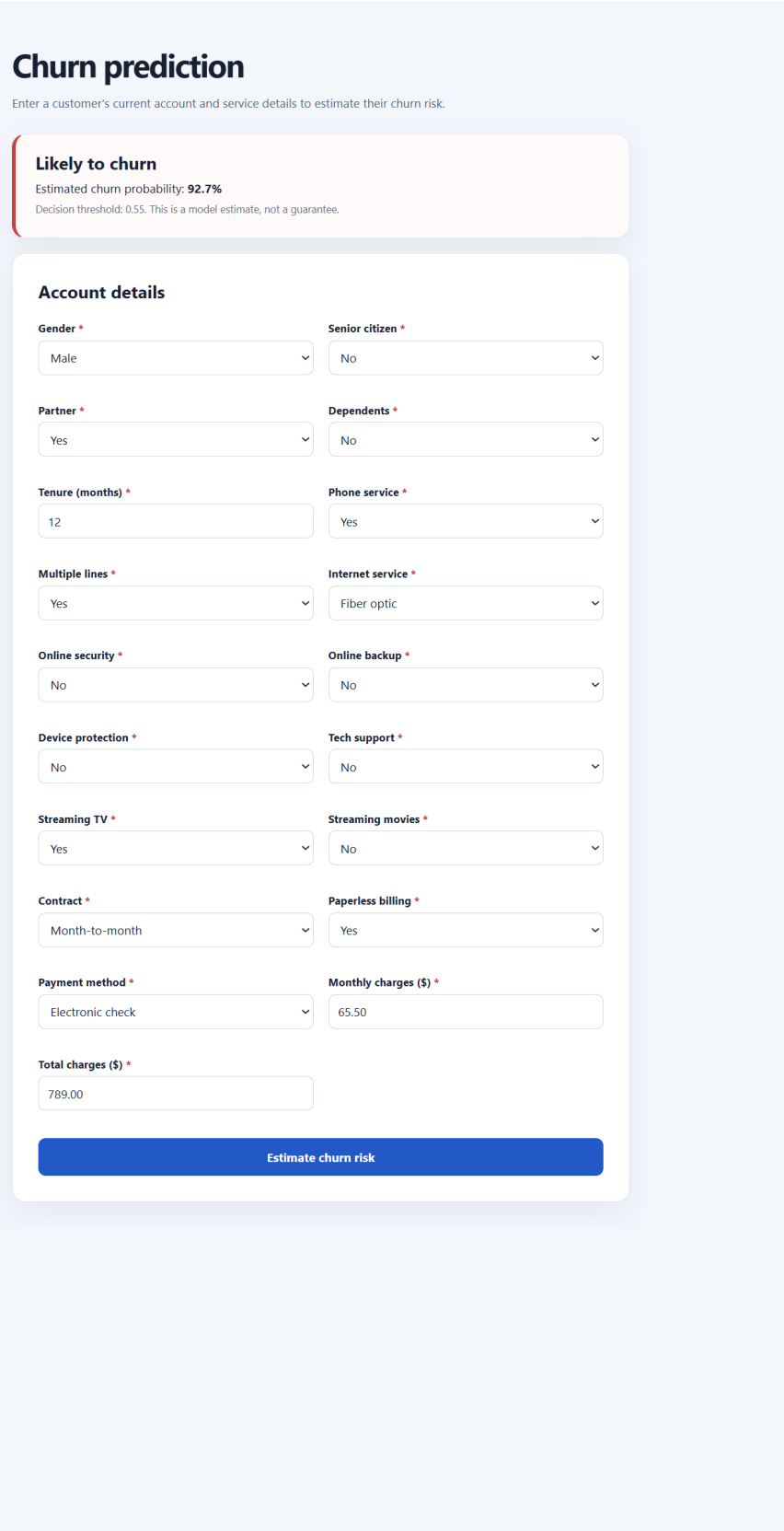
app.py validates the customer inputs and passes a DataFrame in the model feature order to the saved pipeline. The responsive form marks each required field with a red asterisk; native required validation and server-side validation handle incomplete or invalid values.

Seven Flask unit tests cover homepage rendering, expected churn/no-churn examples, missing fields, out-of-range numbers, unexpected categories, and phone-service consistency. Browser submissions confirmed the churn and stay examples.

Run locally: install requirements.txt, run the model-export cell in churn_prediction.ipynb, execute python app.py, and open http://127.0.0.1:5000.

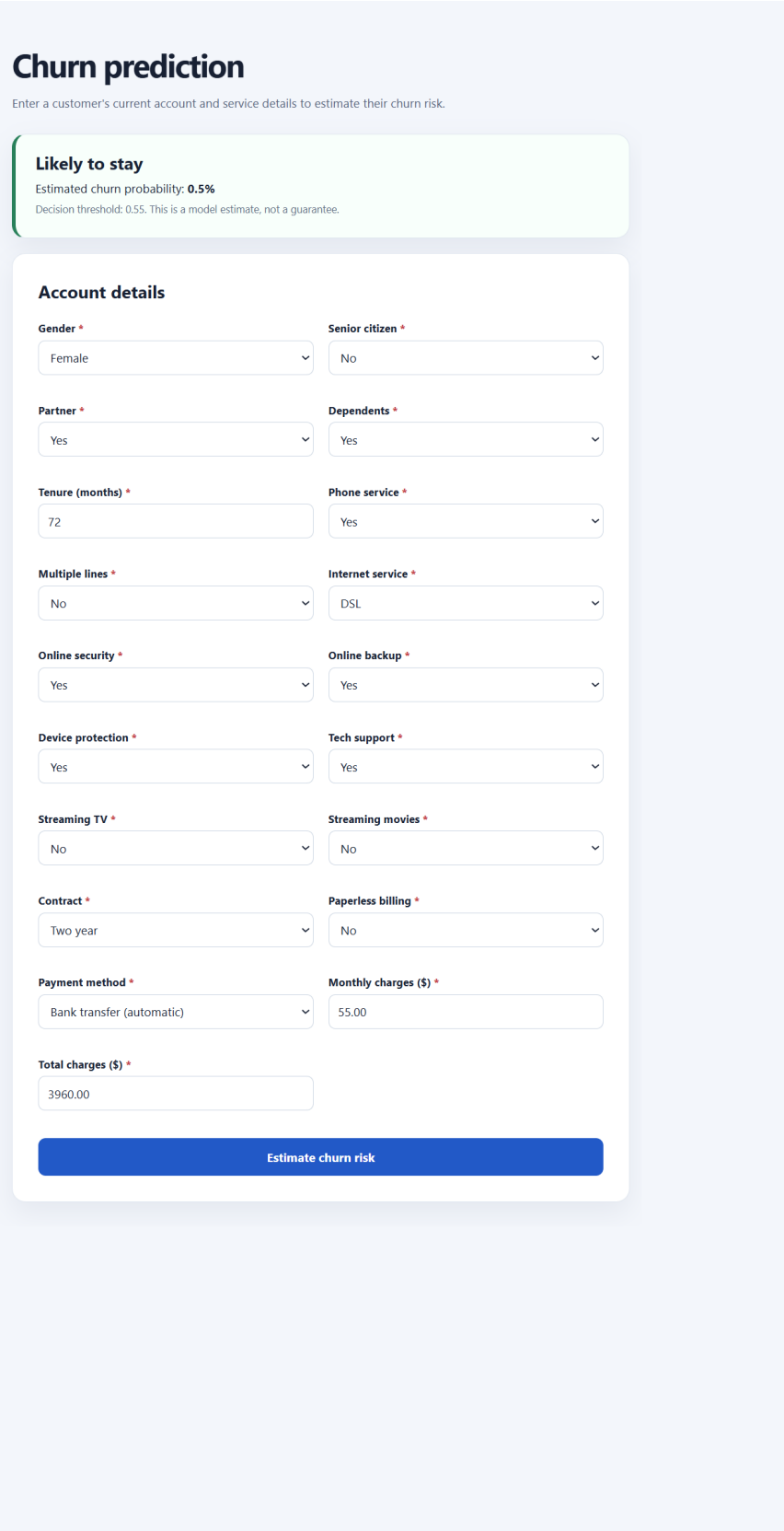
PythonAnywhere deployment remains outstanding until the account owner configures WSGI and provides the live URL. Upload the model and app, install pinned requirements, configure WSGI, reload, and test from another device. The public URL and deployment screenshots should be added after deployment.

Browser verification — churn



Short-tenure, month-to-month fiber example: likely to churn (92.7%).

Browser verification — stay



Long-tenure customer with a two-year contract: likely to stay (0.5% churn probability).